

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 — SUB-STRAND 2.5 RATES OF REACTIONS: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Welcome to your Final Explanation! On Day 1 you watched the Disappearing Thiosulfate Cross: two beakers with the same sodium thiosulfate and dilute hydrochloric acid, one in cold water (about 10 °C, like a cool Nairobi highland morning) and one in warm water (about 50 °C, like a hot Mombasa or Kisumu afternoon). The cross under the warm beaker disappeared in about 30-40 seconds, while the cross under the cold beaker stayed visible for several minutes. Now you will use everything you learned in Lessons 1-8 to explain what happened. Write your answers in the space under each prompt. Use particle ideas (collisions, energy, activation energy) and evidence from our experiments. Use full sentences, and label any diagrams you draw. Think about our driving question: Why do some reactions happen in a flash while others take days, and how can we control their speed in Kenya?

Part 1: Explaining the Disappearing Cross

Look back at the two beakers from Day 1. Both held the same chemicals in the same amounts. (a) How did you measure the speed of the reaction, and what is meant by 'rate of reaction'? (b) Use ideas about particles, collisions and energy to explain why the cross under the warm beaker (about 50 °C) disappeared in about 30-40 seconds while the cross under the cold beaker (about 10 °C) stayed visible for several minutes. Include a labelled particle sketch of both beakers.

Write your answer here:

Part 2: Concentration and Surface Area

In Lessons 3-5 you investigated concentration (diluting sodium thiosulfate to 100%, 75%, 50% and 25%) and surface area (large marble chips, small chips and powder with dilute HCl). (a) Describe the pattern you found for each factor, with the evidence that supports it. (b) Explain each pattern using collision ideas. (c) In the marble experiment all three flasks finally lost the same total mass. What does this tell you about what surface area changes and what it does not change?

Write your answer here:

Part 3: Catalysts and Light

In Lessons 6 and 7 you tested fresh liver, boiled liver and sand with hydrogen peroxide, and you tested manganese dioxide (MnO_2) and light on reactions. (a) What did you observe in these tests, and what evidence showed that MnO_2 is a catalyst? (b) Explain how a catalyst speeds up a reaction, and say how this is different from the way concentration and surface area work. (c) Why did boiled liver not work? (d) How does light speed up some reactions?

Write your answer here:

Part 4: Dagaa Drying at Lake Victoria

A fisher at Lake Victoria wants to dry dagaa and stop it from spoiling quickly. Spoilage is caused by chemical and biological reactions. (a) Choose two rate factors you studied and say how you would change each one to slow down spoilage reactions in the dagaa before it is dried, or to control the reactions. (b) Explain each choice using collision theory. (c) Give one reason why a rate factor that is helpful in one Kenyan situation may be unhelpful in another, using an example such as cooking githeri or the stomach digesting ugali.

Write your answer here:

Part 5: Closing the Case

Write a final answer to our driving question: Why do some chemical reactions happen in a flash while others take days, and how can we control the speed of reactions to make life in Kenya safer, healthier and more productive? In your answer, (a) name all the rate factors we studied and say what each changes at the particle level, and (b) give two examples from Kenyan life where controlling reaction speed is useful.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of collision theory and particle ideas	Consistently explains rate differences using collision frequency, collision energy and activation energy, with clear labelled particle sketches.	Uses collision ideas correctly in most answers but may omit energy or activation energy, or sketches lack some labels.	Gives descriptions of what happened with little or no particle-level explanation, or confuses speed and number of particles.
Use of evidence from the lessons	Accurately uses observations from the thiosulfate cross, concentration, marble and catalyst experiments, including the 30-40 second and several-minute results.	Refers to most experiments with mostly accurate details.	Gives few or inaccurate references to the class experiments.
Distinguishing between rate factors	Clearly separates how temperature, concentration, surface area, catalysts and light each work, including that surface area changes rate but not total product and that catalysts are not consumed.	Explains most factors correctly but with some overlap or missing detail.	Treats the factors as the same or makes incorrect statements about them.
Application to Kenyan contexts	Applies the model accurately and creatively to dagaa drying, githeri, ugali digestion, posho milling or similar, and recognises that factors can be helpful or unhelpful.	Gives relevant Kenyan examples with a reasonable explanation.	Examples are missing, unclear or not linked to the science.

Scientific communication	Writes clear, well-organised answers using correct terms such as rate, activation energy, catalyst and enzyme.	Writes understandable answers with mostly correct vocabulary.	Answers are unclear or use few scientific terms.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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